Question 1

Search the system to identify every file related to rsyslog.

Approach

Use file indexing search tool to quickly locate rsyslog-related files.

Commands

- locate → fast file search using database
- updatedb → update locate database

Output

STEP 1 : update file database

```
┌──(aman㉿kali)-[~]
└─$ sudo updatedb
[sudo] password for aman:

┌──(aman㉿kali)-[~]
└─$
```

STEP 2 : find rsyslog files

```
┌──(aman㉿kali)-[~]
└─$ locate rsyslog
/usr/share/metasploit-framework/modules/auxiliary/dos/syslog/rsyslog_long_tag.rb
```

Key Observations

locate searches from indexed database, not live scan.
rsyslog files usually exist in /etc and /var/log directories.

What I learned

Fast file searching helps identify log storage locations quickly.

Question 2

Modify log configuration so logs are rotated every 7 days.

Approach

Open rsyslog configuration file and change rotation policy to weekly.

Commands

- `nano` → terminal text editor
- `vim` → advanced editor
- `cat` → view file contents

Output

STEP 1 : open configuration file

```
┌──(aman㉿kali)-[~]
└─$ sudo nano /etc/logrotate.conf
```

```
  GNU nano 8.7.1
# see "man logrotate" for details

# global options do not affect preceding include directives

# rotate log files weekly
weekly

# keep 4 weeks worth of backlogs
rotate 4

# create new (empty) log files after rotating old ones
create

# use date as a suffix of the rotated file
#dateext

# uncomment this if you want your log files compressed
#compress

# packages drop log rotation information into this directory
include /etc/logrotate.d

# system-specific logs may also be configured here.
```

STEP 2 : set weekly rotation

```
# rotate log files weekly
weekly
```

Key Observations

Log rotation prevents log files from growing too large.
Rotation frequency affects forensic data availability.

What I learned

Log retention policies control how long system evidence exists.

Question 3

Turn off logging service and observe entries generated in syslog file.

Approach

Stop logging daemon and monitor syslog behavior.

Commands

- `systemctl` → manage system services
- `cat` → read file contents
- `tail` → monitor log updates

Output

STEP 1 : stop logging service

```
┌──(aman㉿kali)-[/]
└─$ sudo systemctl stop rsyslog
Stopping 'rsyslog.service', but its triggering units are still active:
syslog.socket
```

STEP 2 : check syslog activity

```
┌──(aman㉿kali)-[/]
└─$ cat /var/log/syslog
2026-04-02T23:18:18.900870+05:30 kali systemd[1]: Listening on syslog.socket - Syslog Socket.
2026-04-02T23:18:18.900932+05:30 kali systemd[1]: Starting rsyslog.service - System Logging Service...
2026-04-02T23:18:18.904960+05:30 kali systemd[1]: Started rsyslog.service - System Logging Service.
2026-04-02T23:18:18.905981+05:30 kali kernel: Linux version 6.18.12+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-15 (Debian 15.2.0-13) 15.2.0, GNU ld (GNU Binutils for Debian) 2.46)
-02-25)
2026-04-02T23:18:18.905995+05:30 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.18.12+kali-amd64 root=UUID=6fbf4013-33f5-4223-830a-92696c84629c ro quiet splash
2026-04-02T23:18:18.905997+05:30 kali kernel: [Firmware Bug]: TSC doesn't count with P0 frequency!
2026-04-02T23:18:18.905997+05:30 kali kernel: BIOS-provided physical RAM map:
2026-04-02T23:18:18.905998+05:30 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
2026-04-02T23:18:18.905999+05:30 kali kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
2026-04-02T23:18:18.906000+05:30 kali kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
2026-04-02T23:18:18.906002+05:30 kali kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000dffeffff] usable
2026-04-02T23:18:18.906003+05:30 kali kernel: BIOS-e820: [mem 0x00000000dfff0000-0x00000000dfffffff] ACPI data
2026-04-02T23:18:18.906003+05:30 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
2026-04-02T23:18:18.906004+05:30 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
2026-04-02T23:18:18.906005+05:30 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
2026-04-02T23:18:18.906006+05:30 kali kernel: BIOS-e820: [mem 0x0000000100000000-0x000000021fffffff] usable
2026-04-02T23:18:18.906008+05:30 kali kernel: NX (Execute Disable) protection: active
2026-04-02T23:18:18.906009+05:30 kali kernel: APIC: Static calls initialized
2026-04-02T23:18:18.906009+05:30 kali kernel: SMBIOS 2.5 present.
2026-04-02T23:18:18.906010+05:30 kali kernel: DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006
2026-04-02T23:18:18.906011+05:30 kali kernel: DMI: Memory slots populated: 0/0
2026-04-02T23:18:18.906011+05:30 kali kernel: Hypervisor detected: KVM
2026-04-02T23:18:18.906012+05:30 kali kernel: last_pfn = 0x e0000 max_arch_pfn = 0x400000000
2026-04-02T23:18:18.906014+05:30 kali kernel: kvm-clock: Using msrs 4b564d01 and 4b564d00
2026-04-02T23:18:18.906015+05:30 kali kernel: kvm-clock: using sched offset of 14360221055 cycles
2026-04-02T23:18:18.906016+05:30 kali kernel: clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
2026-04-02T23:18:18.906016+05:30 kali kernel: tsc: Detected 2595.126 MHz processor
2026-04-02T23:18:18.906017+05:30 kali kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
2026-04-02T23:18:18.906018+05:30 kali kernel: e820: remove [mem 0x000a0000-0x000fffff] usable
2026-04-02T23:18:18.906023+05:30 kali kernel: last_pfn = 0x220000 max_arch_pfn = 0x400000000
2026-04-02T23:18:18.906024+05:30 kali kernel: MTRR map: 6 entries (3 fixed + 3 variable; max 35), built from 16 variable MTRRs
2026-04-02T23:18:18.906025+05:30 kali kernel: x86/PAT: Configuration [0-7]: WB  WC  UC- UC  WB  WP  UC- WT
2026-04-02T23:18:18.906026+05:30 kali kernel: e820: update [mem 0xe0000000-0xffffffff] usable ==> reserved
2026-04-02T23:18:18.906026+05:30 kali kernel: last_pfn = 0x e0000 max_arch_pfn = 0x400000000
2026-04-02T23:18:18.906027+05:30 kali kernel: found SMP MP-table at [mem 0x0009fbf0-0x0009fbff]
2026-04-02T23:18:18.906029+05:30 kali kernel: RAMDISK: [mem 0x1f37f000-0x2b9b6fff]
2026-04-02T23:18:18.906030+05:30 kali kernel: ACPI: Early table checksum verification disabled
2026-04-02T23:18:18.906030+05:30 kali kernel: ACPI: RSDP 0x00000000000E0000 000024 (v02 VBOX  )
2026-04-02T23:18:18.906031+05:30 kali kernel: ACPI: XSDT 0x00000000DFFF0030 00003C (v01 VBOX   VBOXXSDT 00000001 ASL  00000061)
2026-04-02T23:18:18.906032+05:30 kali kernel: ACPI: FACP 0x00000000DFFF00F0 0000F4 (v04 VBOX   VBOXFACP 00000001 ASL  00000061)
2026-04-02T23:18:18.906032+05:30 kali kernel: ACPI: DSDT 0x00000000DFFF02E0 002353 (v02 VBOX   VBOXBIOS 00000002 VBOX 00029BF3)
2026-04-02T23:18:18.906033+05:30 kali kernel: ACPI: FACS 0x00000000DFFF0200 000040
2026-04-02T23:18:18.906035+05:30 kali kernel: ACPI: FACS 0x00000000DFFF0200 000040
2026-04-02T23:18:18.906036+05:30 kali kernel: ACPI: APIC 0x00000000DFFF0240 000054 (v02 VBOX   VBOXAPIC 00000001 ASL  00000061)
2026-04-02T23:18:18.906036+05:30 kali kernel: ACPI: SSDT 0x00000000DFFF02A0 000038 (v01 VBOX   VBOXCPUT 00000002 VBOX 00029BF3)
2026-04-02T23:18:18.906037+05:30 kali kernel: ACPI: Reserving FACP table memory at [mem 0xdfff00f0-0xdfff01e3]
2026-04-02T23:18:18.906038+05:30 kali kernel: ACPI: Reserving DSDT table memory at [mem 0xdfff02e0-0xdfff2632]
```

STEP 3 : monitor log changes

```
┌──(aman㉿kali)-[/]
└─$ tail /var/log/syslog
2026-04-02T23:18:18.907194+05:30 kali rsyslogd: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.  [v8.2512.0]
2026-04-02T23:18:18.907334+05:30 kali rsyslogd: [origin software="rsyslogd" swVersion="8.2512.0" x-pid="31500" x-info="https://www.rsyslog.com"] start
2026-04-02T23:18:48.218758+05:30 kali systemd[1]: Stopping rsyslog.service - System Logging Service...
2026-04-02T23:18:48.221771+05:30 kali rsyslogd: [origin software="rsyslogd" swVersion="8.2512.0" x-pid="31500" x-info="https://www.rsyslog.com"] exiting on signal 15.
2026-04-02T23:18:48.223545+05:30 kali systemd[1]: rsyslog.service: Deactivated successfully.
2026-04-02T23:18:48.224639+05:30 kali systemd[1]: Stopped rsyslog.service - System Logging Service.
2026-04-02T23:18:48.263469+05:30 kali systemd[1]: Starting rsyslog.service - System Logging Service...
2026-04-02T23:18:48.407026+05:30 kali systemd[1]: Started rsyslog.service - System Logging Service.
2026-04-02T23:18:48.409128+05:30 kali rsyslogd: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.  [v8.2512.0]
2026-04-02T23:18:48.410581+05:30 kali rsyslogd: [origin software="rsyslogd" swVersion="8.2512.0" x-pid="31758" x-info="https://www.rsyslog.com"] start
```

Key Observations

Stopping rsyslog reduces logging events.
System may record service stop event before logging halts.

What I learned

Disabling logs can reduce traces of system activity.

Question 4

Securely overwrite and remove kernel log files permanently.

Approach

Use secure deletion tool that overwrites data before removing files.

Commands

- `shred` → overwrite file data
- `rm` → remove files
- `Is` → verify deletion

Output

STEP 1 : locate kern log files

```
┌──(aman㉿kali)-[/]
└─$ ls /var/log/kern*
/var/log/kern.log
```

STEP 2 : securely delete logs

```
┌──(aman㉿kali)-[/]
└─$ sudo shred -u /var/log/kern.log
```

Key Observations

shred overwrites file multiple times.
Recovery becomes extremely difficult after shredding.

What I learned

Secure deletion prevents forensic recovery of sensitive logs.

Security Insights

1. Location log files helps attackers identify evidence storage paths.
2. Log rotation settings influence how long activity traces remain.
3. Disabling logging reduces detection visibility of malicious actions.
4. Secure deletion techniques prevent recovery of incriminating logs.